A SYSTEM FOR COVERT VISUAL DATA EXFILTRATION VIA A HARDWARE-MODIFIED DISPLAY DEVICE

Title: A SYSTEM FOR COVERT VISUAL DATA EXFILTRATION VIA A HARDWARE-MODIFIED DISPLAY DEVICE

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SPECIFICATION

Abstract

A system for the covert surveillance of a user's computer activity. The system comprises a computer display monitor that has been physically modified at the hardware level to include a covert data exfiltration module. This module is embedded within the monitor's housing and is positioned to intercept the raw video signal before it is rendered on the screen. The module captures the video frame data, compresses it, and transmits it wirelessly to a remote receiver. Because the surveillance mechanism is entirely contained within the hardware of the display device, it is undetectable by any software-based security or anti-malware running on the host computer. The system allows for the complete, real-time capture of a target's visual data without their knowledge.

Background of the Invention

The monitoring of computer activity for intelligence or security purposes is a well-established field. The majority of methods rely on software, such as keyloggers, screen recorders, or remote access trojans (RATs), installed on the target computer's operating system. However, these software-based methods are vulnerable to detection and removal by increasingly sophisticated anti-virus and anti-malware products. As a result, they may have a limited operational lifespan.

Hardware-based attacks, often called "hardware Trojans," offer a more persistent and stealthy alternative. Prior art has explored devices that intercept data from keyboard cables (hardware keyloggers) or from the video output port of a computer. However, these

external "man-in-the-middle" devices are physically visible and can be discovered by a visual inspection. There is a need for a hardware-based screen-capture system that is completely concealed from the user.

Summary of the Invention

The present invention is a hardware-based surveillance system that is fully integrated and concealed within the housing of a standard computer monitor. The system provides for the undetectable capture and transmission of a user's screen content.

The system is created by modifying a conventional computer monitor. A custom-designed hardware module is embedded within the monitor's internal chassis. This "exfiltration module" comprises a video signal tap, a small processor with a frame buffer, and a wireless transmitter.

The video signal tap is physically connected to the internal wiring that carries the video signal from the monitor's input port (e.g., HDMI, DisplayPort) to its display controller board. This allows the module to intercept a raw, unencrypted copy of the video data intended for the screen.

The intercepted frame data is temporarily stored in the frame buffer and then processed by the onboard processor. The processor can be configured to capture still frames at a set interval or to compress a continuous video stream. The resulting data is then transmitted wirelessly via the onboard transmitter to a receiver located in the vicinity of the target.

Because the entire system is self-contained within the monitor's housing and intercepts the video signal at the hardware level, it is completely invisible to the host computer's operating system and any security software it may be running. The target user perceives only a normal, functioning monitor, while their every action is being visually recorded and exfiltrated.

Brief Description of the Drawings

FIG. 1 is a system diagram illustrating the data flow from the host computer to the remote receiver via the modified monitor.

FIG. 2 is a block diagram showing the internal components of the modified monitor and the embedded exfiltration module.

Detailed Description of the Invention

Referring to **FIG. 1**, the system comprises a host computer (101) which generates a standard video signal (102). This signal is sent to a seemingly ordinary computer monitor, the Modified Display Device (103). The monitor displays the image to the user as expected.

However, concealed within the Modified Display Device (103), an embedded hardware module intercepts the video signal. This module captures the visual data and then wirelessly transmits it (104) to a remote Receiver (105). The host computer (101) has no awareness of this secondary data path, as its interaction with the monitor is identical to that of a standard, unmodified device.

FIG. 2 details the internal components of the Modified Display Device. The monitor housing (201) contains the standard components of a monitor, including a video input port (202) and a display controller board (203) which drives the physical screen (204). The video signal path (205) normally runs directly from the input port to the controller board.

In the modified device, a hardware-based Exfiltration Module (206) is embedded within the housing. This module includes a Video Signal Tap (207) which is physically spliced into the video signal path (205). The tap duplicates the video signal, sending one copy to the standard display controller (203) so the monitor functions normally, and a second copy to the exfiltration module.

The signal entering the exfiltration module is captured by a Frame Grabber/Processor (208). This component captures and optionally compresses the video data, which is then passed to a Wireless Transmitter (209). The transmitter sends the captured screen data via an antenna (210) to a remote receiver. The entire exfiltration module is self-contained and does not interact with the host computer in any way that can be detected by software.

~~Claims~~

~~What is claimed is:~~

1. A system for covert data surveillance, comprising:
    - a computer display monitor having a housing;
    - a data exfiltration module physically embedded and concealed entirely within said housing;
    - wherein the data exfiltration module comprises:
        - a video signal tap connected to an internal video signal path of the monitor;

- a processor configured to receive captured video data from the signal tap; and
- a wireless transmitter configured to transmit the captured video data to a remote receiver.

2. The system of claim 1, wherein the video signal tap is configured to duplicate the video signal, allowing one copy to proceed to the monitor's display controller while the second copy is sent to the processor.

3. The system of claim 1, wherein the processor is further configured to compress the captured video data before transmission.

4. The system of claim 1, wherein the system is undetectable by software running on a host computer connected to the computer display monitor.

5. A method for covertly capturing visual data from a computer, comprising the steps of:

   - physically modifying the internal hardware of a computer monitor to install a self-contained exfiltration module;
   - intercepting, via the exfiltration module, the internal video signal being transmitted from the monitor's input port to its display controller;
   - capturing the frame data of the intercepted video signal; and
   - wirelessly transmitting the captured frame data from the exfiltration module to a remote receiving device.

6. The method of claim 5, wherein the host computer connected to the modified monitor is not aware of the interception and transmission of the frame data.

7. The method of claim 5, further comprising the step of compressing the captured frame data within the exfiltration module prior to transmission.

Drawings

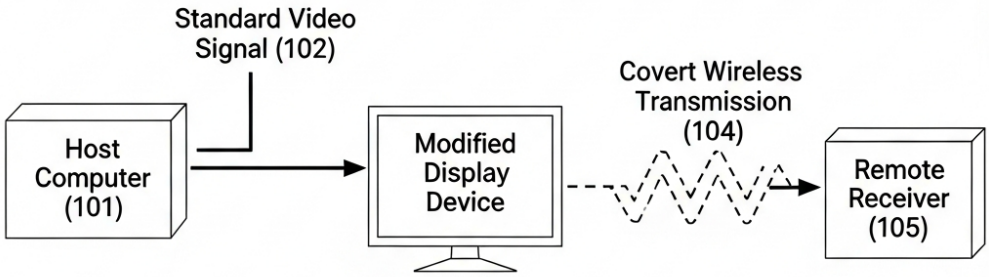


Figure 1: System data flow diagram.

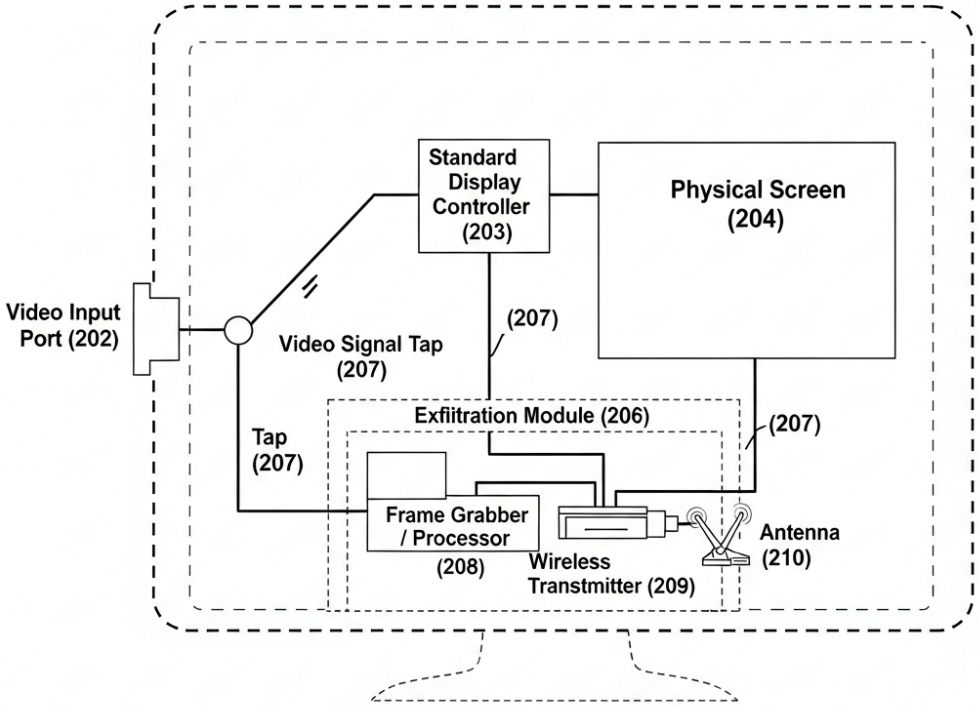


Figure 2: Internal block diagram of the modified monitor.